



Andy Casafranca.

Mechatronics Engineer ·
Embedded Systems & AI

01 PROFILE

Mechatronics Engineering student with hands-on experience designing **embedded systems**, tuning **AI models**, and delivering end-to-end IoT solutions. Combines a strong electronics and firmware foundation with software-engineering practice and bilingual, client-facing skills. Driven by precision, reliability, and shipping working hardware.

02 EXPERIENCE

Feb 2025 — Jun 2025 Lima, Perú

REMOTE

Bilingual Customer Support Agent Major U.S. last-mile delivery platform

- Supported delivery drivers for a leading U.S. last-mile delivery platform, resolving order, routing, and payment issues in real time over chat and phone.
- Handled a high volume of bilingual (English/Spanish) interactions while consistently meeting strict response-time and quality targets.
- De-escalated time-sensitive incidents and coordinated fast resolutions to keep deliveries on track.

EN / ES Real-time support

Logistics CRM tooling

2023 — 2024 Lima, Perú

RESEARCH

Undergraduate Research Assistant

Universidad Peruana de Ciencias Aplicadas (UPC)

- Designed a **custom analog front-end** for an EEG/EMG acquisition headset and developed the STM32-based firmware for real-time signal capture.
- Built a Python signal-processing pipeline (NumPy, SciPy) achieving **87% classification accuracy** on motor-imagery tasks across 8 channels.

STM32 Python SciPy MATLAB

Analog design

Feb 2023 — Aug 2023 Lima, Perú

ON-SITE

Bilingual Customer Support Agent

Major U.S. health-insurance network

- Assisted U.S. healthcare providers with claims, eligibility, and benefits inquiries on behalf of a major U.S. health-insurance network, under strict regulatory protocols.
- Delivered accurate bilingual (English/Spanish) support in a regulated, high-volume on-site call-center environment.

EN / ES Healthcare

Regulated environment

Claims & eligibility

2022 — 2023 Remote

FREELANCE

Embedded Systems & Automation Engineer

Independent projects

- Delivered three turnkey IoT projects end-to-end — hardware, firmware, and dashboards — including a smart-greenhouse mesh and a water-quality monitor.
- Authored ESP32 mesh-networking firmware in C/C++ with OTA updates and integrated React dashboards over WebSocket.
- Performed a responsible-disclosure security assessment of a consumer IoT camera, resulting in a **CVE writeup** with proof-of-concept exploit.

ESP32 C++ MQTT React

Burp Suite

03 SELECTED PROJECTS

Hackatón MINSA 2017 1ST PLACE

Ministry of Health · Team Vincere

Part of the 10-student team “Vincere” that won first place at Peru's national MINSA health hackathon, building a video-game-based app for preliminary depression screening.

Healthtech · Mental health · Game-based screening

HydraDetect-AI IN PROGRESS

Clandestine water-tap detection

Detects clandestine water connections in city pipes using a controlled water hammer and machine learning — 89% field accuracy on an offline Arduino edge device.

Arduino · Python · Machine Learning · Edge AI

04 EDUCATION

2019 — Present Expected 2026

BACHELOR

B.Sc. in Mechatronics Engineering

Universidad Peruana de Ciencias Aplicadas (UPC) · Lima, Perú

- Focus areas: **control systems**, **embedded firmware**, **biomedical instrumentation**, and **applied AI**.
- Selected coursework: Digital Signal Processing · Control Systems · Microprocessors · Robotics · Machine Learning · Bioengineering.

2024 Online

CERT.

Practical Ethical Hacking & Network Security

Self-directed · TCM Security · TryHackMe

- Hands-on penetration-testing labs covering OWASP Top-10 and active-directory exploitation paths.

CONTACT

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TECHNICAL SKILLS

Languages

Python C C++ C#
JavaScript ASM MATLAB

Embedded · Hardware

STM32 ESP32 Arduino RTOS
PWM I²C UART PCB design
QMK

AI · Signals

PyTorch TensorFlow OpenCV
DSP FFT EEG/EMG PID

Tools · CAD

SolidWorks Fusion 360 KiCad
Git Linux Docker

Security

Burp Suite Nmap Wireshark
Metasploit

Frontend

React WebSockets HTML/CSS
Figma

LANGUAGES

Spanish NATIVE

English PROFESSIONAL · C1

INTERESTS

Formula 1 Road trips
Ocean Gaming
Photography